

LaTeX Recipes for a Master's Thesis

Copy-paste templates (figures, tables, links, footnotes, macros, long equations, and more) with explanations of key commands and parameters.

How to use this PDF

Each section contains: (1) a template snippet you can paste into your thesis, and (2) a concise explanation of what the important commands and options do. These recipes are intentionally generic so they work with most university thesis classes.

0. Preamble essentials (packages you will reuse)

```
% --- core thesis utilities ---
\usepackage[T1]{fontenc}      % better font encoding for PDF output
\usepackage{lmodern}          % Latin Modern fonts (good default)
\usepackage{microtype}        % subtle typographic improvements

% --- math ---
\usepackage{amsmath, amssymb} % align/split + extra symbols
\usepackage{mathtools}        % extensions to amsmath (e.g., \coloneqq)

% --- figures & tables ---
\usepackage{graphicx}          % \includegraphics
\usepackage{caption}           % caption control
\usepackage{subcaption}        % subfigures
\usepackage{booktabs}          % \toprule \midrule \bottomrule
\usepackage{siunitx}           % aligned numbers + units in tables

% --- references & links ---
\usepackage[hidelinks]{hyperref} % clickable refs; hidelinks removes colored boxes
\usepackage[nameinlink]{cleveref} % "Figure", "Eq.", etc. automatically
\usepackage{xurl}              % better URL line breaking
```

Notes: Load **hyperref** late (after most packages). **cleveref** builds on labels/refs to print context-aware references (Figure vs Equation) and keeps them clickable.

1. Figure with caption, label, and reference

Template

```
\begin{figure}[tbp]
  \centering
  \includegraphics[width=0.85\textwidth]{figures/my_plot.pdf}
  \caption{Transit depth as a function of orbital period for the sample.}
  \label{fig:depth_vs_period}
\end{figure}
```

As shown in `\cref{fig:depth_vs_period}`, ...

Explanation

- `\begin{figure}...\end{figure}` creates a float (LaTeX may move it to improve layout).

- `[tbp]` requests placement at top (t), bottom (b), or a float-only page (p). It is a preference, not a guarantee.
- `\centering` centers contents inside the float.
- `\includegraphics[width=...]{...}` (graphicx) inserts an image; `width=0.85\textwidth` scales it to 85% of the text block width.
- `\caption{...}` prints the caption and assigns the figure number.
- `\label{...}` defines the reference anchor. Put it after `\caption` so it captures the correct number.
- `\cref{...}` (cleveref) prints the object type + number (e.g., Figure 2) and hyperlinks it in the PDF.

2. Two-panel figure with subcaptions (a) and (b)

Template

```
\begin{figure}[tbp]
  \centering

  \begin{subfigure}[t]{0.48\textwidth}
    \centering
    \includegraphics[width=\textwidth]{figures/panel_a.pdf}
    \caption{Raw light curve.}
    \label{fig:lc_raw}
  \end{subfigure}
  \hfill
  \begin{subfigure}[t]{0.48\textwidth}
    \centering
    \includegraphics[width=\textwidth]{figures/panel_b.pdf}
    \caption{Detrended light curve.}
    \label{fig:lc_detrended}
  \end{subfigure}

  \caption{Photometry before and after detrending.}
  \label{fig:lc_comparison}
\end{figure}
```

See `\cref{fig:lc_comparison}` and `\cref{fig:lc_raw,fig:lc_detrended}`.

Explanation

- `subfigure` (subcaption) makes labeled panels like Figure 3a and 3b.
- `[t]` top-aligns the subfigures; alternatives include `[b]` bottom or `[c]` center.
- `{0.48\textwidth}` sets each panel width so two fit on one line; tweak to taste.
- `\hfill` inserts flexible horizontal space between panels.
- Each subfigure can have its own `\caption` and `\label`; the overall figure also has a `\caption` and `\label`.
- `\cref{fig:lc_raw,fig:lc_detrended}` references both panels in one go.

3. Table with caption and label (booktabs style)

Template

```

\begin{table}[tbp]
  \centering
  \caption{Summary of the cleaned sample.}
  \label{tab:sample_summary}

  \begin{tabular}{lrr}
    \toprule
    Category & Count & Fraction \\
    \midrule
    M dwarfs & 1200 & 0.80 \\
    K dwarfs & 250 & 0.17 \\
    Other    & 50 & 0.03 \\
    \bottomrule
  \end{tabular}
\end{table}

```

See `\cref{tab:sample_summary}`.

Explanation

- `table` is a float (like `figure`); `[tbp]` works the same way.
- `tabular` defines the grid; `{lrr}` means 3 columns: left, right, right aligned.
- `booktabs` provides `\toprule`, `\midrule`, `\bottomrule` for professional-looking horizontal rules.
- Avoid vertical rules (`|`) unless your style guide forces them; `booktabs` is designed for cleaner tables.

4. Tables with aligned decimals and units (`siunitx`)

Template

```

\sisetup{
  round-mode = places,
  round-precision = 3,
}

\begin{table}[tbp]
  \centering
  \caption{Example stellar parameters.}
  \label{tab:stellar_params}

  \begin{tabular}{l S[table-format=1.3] S[table-format=2.1]}
    \toprule
    Star & {Mass ( $M_{\odot}$ )} & {Distance (pc)} \\
    \midrule
    A & 0.5123 & 12.4 \\
    B & 0.2719 & 98.1 \\
    C & 0.8432 & 7.0 \\
    \bottomrule
  \end{tabular}
\end{table}

```

Explanation

- `\sisetup{...}` sets global formatting for `siunitx` numeric columns.
- `round-mode=places` and `round-precision=3` round numbers to 3 decimal places (where applicable).

- S columns parse numbers and align on the decimal marker; table-format hints expected digit counts (e.g., 1.3).
- Wrap S-column headers in braces { ... } so siunitx does not try to parse them as numbers.

5. URL links and clickable text

Template

Data are hosted at `\url{https://archive.stsci.edu/}`.
 The analysis code is available at `\href{https://github.com/user/repo}{GitHub repository}`.

Explanation

- `\url{...}` prints a URL and allows line breaks (xurl improves breaking).
- `\href{target}{text}` makes 'text' clickable and points to 'target'.
- `hyperref` enables clickability in the PDF.

6. Hyperlinked footnotes you can jump back to later

Template

This approach follows the standard reduction pipeline%
`\footnote{\hypertarget{fn:pipeline}{Full details are provided in Appendix~A.}}`.

Later, we refer back to that note (`\hyperlink{fn:pipeline}{see footnote}`).

Explanation

- `\footnote{...}` inserts a numbered footnote.
- % at end of a line prevents an unwanted space that would otherwise appear in the output.
- `\hypertarget{name}{...}` defines an internal anchor at that location.
- `\hyperlink{name}{text}` creates a clickable link to that anchor.
- This is name-based: it stays correct even if footnote numbers change.

7. Custom macros (functions) and symbol shortcuts

Template

% One-argument macro:
`\newcommand{\snr}[1]{\mathrm{SNR}\,, (#1)}`

% Two-argument macro:
`\newcommand{\qty}[2]{#1\,, \mathrm{#2}}`

% Symbol shortcuts:
`\newcommand{\msun}{M_\odot}`
`\newcommand{\rsun}{R_\odot}`

% 'Defined as' operator (mathtools):

```
\newcommand{\defeq}{\coloneqq}
```

Example: $\text{\$}\text{\texttt{snr}}\{\text{\texttt{\lambda}}\} = 25\text{\$}$, $\text{\$}0.3\text{\$,}\text{\texttt{msun}}\text{\$}$, $\text{\$}\text{\texttt{qty}}\{12.4\}\{\text{\texttt{pc}}\}\text{\$}$, and $\text{\$}a\ \text{\texttt{\defeq}}\ b+c\text{\$}$.

Explanation

- `\newcommand{name}[n]{definition}` defines a macro; [n] is the number of arguments (default 0).
- Arguments are referenced as #1, #2, ... inside the definition.
- `\mathrm{...}` typesets upright roman text in math mode (good for operators and units).
- `\coloneqq` (mathtools) is a clean 'defined as' symbol.

8. Macros for text terms with automatic spacing (xspace)

Template

```
\usepackage{xspace}
```

```
\newcommand{\tess}{\textit{TESS}\xspace}
\newcommand{\kepler}{\textit{Kepler}\xspace}
```

We analyze `\tess` data. The `\kepler` mission provides comparison baselines.

Explanation

- `\textit{...}` italicizes in text mode (check your style guide for mission name styling).
- `\xspace` tries to insert a space after the macro when appropriate, so you can write '`\tess data`' naturally.
- Trade-off: `\xspace` can guess wrong in edge cases (punctuation); remove it if it misbehaves.

9. Long equations: align, split, and manual line breaks

Template

```
% Multi-line aligned equations (each line numbered by default)
```

```
\begin{align}
y(t) &= m\,t + b + \epsilon(t), \quad \backslash\backslash
\epsilon(t) &\sim \mathcal{N}(0, \sigma^2).
\end{align}
```

```
% One equation number for multiple lines
```

```
\begin{equation}
\begin{split}
&\mathcal{L}(\theta) \\
&= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_i^2}} \\
&\quad \exp\!\left[-\frac{(y_i - f(x_i;\theta))^2}{2\sigma_i^2}\right] \quad \backslash\backslash \\
&= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_i^2}} \\
&\quad \exp\!\left[-\frac{r_i(\theta)^2}{2\sigma_i^2}\right].
\end{split}
\end{equation}
```

```
% Manual break + suppress numbering on a line
```

```
\begin{align}
F &= A + B + C + D + E + F + G + H \quad \nonumber \quad \backslash\backslash \\
&\quad + I + J + K.
\end{align}
```

```
\end{align}
```

Explanation

- align (amsmath) aligns lines at '&' and gives each line a number unless suppressed.
- Use '\\" to end a line within align/split.
- equation provides a single equation number; split formats multiple lines inside that one number.
- \nonumber suppresses numbering on a specific align line.
- \quad adds indentation; keep a leading '&' on continuation lines to preserve alignment.

10. Clean referencing of equations, figures, and tables

Template

```
\begin{equation}
  E = mc^2
  \label{eq:mass_energy}
\end{equation}
```

Using \cref{eq:mass_energy}, we compute ...

Explanation

- \label{...} stores the current counter value (equation/figure/table) for later reference.
- \cref{...} prints the type + number and hyperlinks it (requires cleveref + hyperref).
- Alternative: \eqref{...} (amsmath) prints only '(1)' without the 'Eq.' prefix.

11. Citations and bibliography with biblatex (author-year)

Template

```
\usepackage[
  backend=biber,
  style=authoryear,
  maxcitenames=2,
  maxbibnames=99,
]{biblatex}
\addbibresource{references.bib}
```

As shown by \textcite{Doe2020}, ...

Other work finds similar trends \parencite{Smith2019,Johnson2021}.

```
\printbibliography
```

Explanation

- biblatex manages citations and the bibliography; backend=biber is recommended for modern workflows.
- style=authoryear yields author-year citations; numeric is another common style.
- maxcitenames controls how many authors appear before 'et al.' in citations.

- `\textcite{key}` produces 'Author (Year)'; `\parencite{key}` produces '(Author Year)'.
- `\printbibliography` prints the reference list.

12. Appendices

Template

```
\appendix
```

```
\chapter{Additional Derivations}
\label{app:derivations}
```

Extra material referenced from the main text.

Explanation

- `\appendix` switches chapter numbering to Appendix A, B, ... (class-dependent).
- `\chapter` creates a new appendix chapter in report/book-like classes.
- `\label` allows referencing the appendix later (e.g., `\cref{app:derivations}`).

13. Float control when figures/tables drift too far

Template

```
\usepackage{placeins} % in preamble
```

```
% ... later in your document ...
\FloatBarrier
```

Explanation

- `\FloatBarrier` prevents floats from moving past that point (useful at end of a section).
- Trade-off: overusing barriers can create awkward whitespace if LaTeX cannot place floats nicely.

14. Force exact placement (use sparingly)

Template

```
\usepackage{float}
```

```
\begin{figure}[H]
  \centering
  \includegraphics[width=\textwidth]{figures/x.pdf}
  \caption{Placed exactly here.}
\end{figure}
```

Explanation

- `[H]` (float package) disables floating and forces placement at that location.
- Trade-off: can harm page layout; reserve for cases where exact placement is essential.

15. Theorem-like environments (definitions, lemmas)

Template

```
\usepackage{amsthm}

\newtheorem{definition}{Definition}
\newtheorem{lemma}{Lemma}

\begin{definition}[Signal-to-noise ratio]
Let  $S$  be the signal and  $\sigma$  the noise. Then  $\mathrm{SNR} \stackrel{\text{def}}{=} S/\sigma$ .
\end{definition}
```

Explanation

- `amsthm` provides structured theorem environments with consistent formatting and numbering.
- `\newtheorem{definition}{Definition}` creates a 'definition' environment labeled 'Definition'.
- Optional `[Title]` adds a descriptor after the label (e.g., 'Definition 1 (Signal-to-noise ratio)').

16. TODO notes you can remove cleanly before submission

Template

```
\usepackage{xcolor}

\newcommand{\todo}[1]{\textcolor{red}{[TODO: #1]}}

\todo{Cite the most recent review here.}

% Before final submission, you can disable TODOs:
\renewcommand{\todo}[1]{}
```

Explanation

- `\textcolor{red}{...}` (`xcolor`) colors text; you can choose other colors if needed.
- `\todo{...}` is a wrapper macro to keep TODOs consistent and searchable.
- `\renewcommand{\todo}[1]{}` redefines the macro to print nothing (hides all TODOs).

Quick checklist before you submit

- Run a clean build: remove auxiliary files (`.aux`, `.bbl`, `.bcf`, `.blg`, `.run.xml`) and recompile.
- Verify every `\label` has a corresponding `\cref`/`\ref` and resolves correctly (no '??').
- Check that URLs break nicely and do not overflow margins (`xurl` helps).
- Check float placement: if figures drift too far, try `\FloatBarrier` at section boundaries.
- Make sure your department template allows your chosen packages (some universities restrict).